MTH107 2023S2 Exam

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*	MTH219	Fundamentals of Statistics and Probability
*	MTH220	Statistical Methods and Inference
*	MTH355	Basic Mathematical Optimisation

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MTH107

Examination - July Semester 2023

Calculus I

Tuesday, 14 November 2023 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **FOUR (4)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1a

Determine

$$\lim_{x \to 0} \frac{e^{2x} - 1}{e^x - 1}$$

<u>without using L'Hopital's Rule</u>. Show your steps clearly.

(10 marks)

Question 1b

Show that there exists a real root for the equation $e^x = x^2 - 3$. Justify your answer.

(10 marks)

Question 2a

Define the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} (x-2)^2 \sin(\frac{1}{x-2}) & \text{if } x \neq 2, \\ 0 & \text{if } x = 2. \end{cases}$$

Discuss whether f is continuous at $x = 2$.

(10 marks)

Question 2b

Let $f(x) = |x|e^x$.

Discuss whether f is differentiable at 0. If yes, determine the value of $f'(0)$.

(10 marks)

Question 3a

Determine

$$\lim_{x \to 1} \left(\frac{2x}{x-1} - \frac{2}{\ln x} \right).$$

Show your steps clearly.

(10 marks)

Question 3b

Let f be a continuous and differentiable function on $\mathbb{R}$. Suppose that $f(-1) = -8$ and $f'(x) \leq 3$ for all $x \in \mathbb{R}$. Determine the largest possible value of $f(2023)$.

(10 marks)

Question 4a

Suppose that

$$4x^3 - 33x^2 + 92x + 1936 = a_0 + a_1(x-3) + a_2(x-3)^2 + a_3(x-3)^3$$

for all $x \in \mathbb{R}$.

<u>Without solving any simultaneous equations</u>, determine the values of a_0, a_1, a_2, a_3.

(10 marks)

Question 4b

Determine the Taylor expansion of

$$xe^x$$

about $x = 0$, up to and including the term in x^3. Use this Taylor expansion to estimate the value of $0.1e^{0.1}$, giving your answer to 2 decimal places. Estimate the error in your evaluation.

(10 marks)

Question 5

Define the function $g : \mathbb{R} \longrightarrow \mathbb{R}$ by

$$g(x) = \frac{1+x}{2+x^2}.$$

Question 5a

Apply the first derivative test to compute the intervals on which g is increasing.

(10 marks)

Question 5b

Determine the critical point(s) of g. Apply relevant derivative tests and discuss the nature of these critical point(s). Show your steps clearly.

(10 marks)

----- END OF PAPER -----

1(a) $\lim_{x\to 0} \frac{e^{2x}-1}{e^x-1} = \lim_{x\to 0} \frac{(e^x-1)(e^x+1)}{e^x-1}$

$= \lim_{x\to 0} (e^x+1)$

$= 1+1$

$= 2$

(b) $e^x = x^2-3$

$x^2-3-e^x = 0$

let $f(x) = x^2-3-e^x$

$f(0) = 0-3-1 = -4$

$f(-3) = 9-(-3)-e^{-3} = 12-e^{-3}$

Since $f(0) < 0$ and $f(-3) > 0$

By Intermediate Value Theorem,

$\exists c$ on $(-3,0)$ such that $f(c)=0$

$\therefore$ There is a real root for the equation $e^x = x^2-3$

2(a) $-|(x-2)^2| \le -|(x-2)^2||\sin(\frac{1}{x-2})| \le (x-2)^2\sin(\frac{1}{x-2}) \le |(x-2)^2||\sin(\frac{1}{x-2})| \le |(x-2)^2|$

$-|(x-2)^2| \le (x-2)^2\sin(\frac{1}{x-2}) \le |(x-2)^2|$

$\lim_{x\to 2} -|(x-2)^2| \le \lim_{x\to 2} (x-2)^2\sin(\frac{1}{x-2}) \le \lim_{x\to 2} |(x-2)^2|$

$0 \le \lim_{x\to 2} (x-2)^2\sin(\frac{1}{x-2}) \le 0$

By Squeeze Theorem, $\lim_{x\to 2} (x-2)^2\sin(\frac{1}{x-2}) = 0$

$f(2) = 0$

$\lim_{x\to 2} f(x) = \lim_{x\to 2} (x-2)^2\sin(\frac{1}{x-2}) = 0$

Since $\lim_{x\to 2} f(x) = f(2)$, then $f(x)$ is continuous at $x=0$.

(b) $f(x) = \begin{cases} xe^x & \text{if } x \ge 0 \\ -xe^x & \text{if } x < 0 \end{cases}$

$\lim_{h\to 0^+} \frac{f(h)-f(0)}{h} = \lim_{h\to 0^+} \frac{he^h-0}{h} = \lim_{h\to 0^+} e^h = 1$

$\lim_{h\to 0^-} \frac{f(h)-f(0)}{h} = \lim_{h\to 0^-} \frac{-he^h-0}{h} = \lim_{h\to 0} -e^h = -1$

$\Rightarrow \lim_{h\to 0} \frac{f(h)-f(0)}{h}$ doesn't exist

$\therefore f(x)$ is not differentiable at $x=0$.

3(a)

$$\lim_{x\to 1}\left[\frac{2x}{x-1}-\frac{2}{\ln x}\right] = \lim_{x\to 1}\left[\frac{2(x-1)+2}{x-1}-\frac{2}{\ln x}\right]$$

$$= \lim_{x\to 1}\left[2+\frac{2}{x-1}-\frac{2}{\ln x}\right]$$

$$= 2+2\lim_{x\to 1}\left[\frac{1}{x-1}-\frac{1}{\ln x}\right]$$

$$= 2+2\lim_{x\to 1}\left[\frac{\ln x-(x-1)}{(x-1)\ln x}\right]$$ $\frac{0}{0}$ form $\Rightarrow$ L'Hopital Rule

$$= 2+2\lim_{x\to 1}\left[\frac{\frac{1}{x}-1}{\ln x+\frac{x-1}{x}}\right]$$

$$= 2+2\lim_{x\to 1}\left[\frac{\frac{1}{x}-1}{\ln x+1-\frac{1}{x}}\right]$$ $\frac{0}{0}$ form $\Rightarrow$ L'Hopital Rule

$$= 2+2\lim_{x\to 1}\left[\frac{-\frac{1}{x^2}}{\frac{1}{x}+\frac{1}{x^2}}\right]$$

$$= 2+2\left[\frac{-1}{1+1}\right]$$

$$= 2+2\left(-\frac{1}{2}\right)$$

$$= 1$$

3(b) By Mean Value Theorem, $\exists c$ on $(-1, 2023)$ such that.

$$f'(c) = \frac{f(2023)-f(-1)}{2023-(-1)}$$

$$= \frac{f(2023)-(-8)}{2024}$$

$$= \frac{f(2023)+8}{2024}$$

since $f'(x) \leq 3$

$f'(c) \leq 3$

$$\frac{f(2023)+8}{2024} \leq 3$$

$f(2023)+8 \leq 6072$

$\therefore f(2023) \leq 6064$

largest possible value of $f(2023) = 6064$

4(a) $f(x) = 4x^3 - 33x^2 + 92x + 1936$

$f'(x) = 12x^2 - 66x + 92$

$f''(x) = 24x - 66$

$f'''(x) = 24$

$a_0 + a_1(x-3) + a_2(x-3)^2 + a_3(x-3)^3$ is just Taylor Expansion of $f(x)$ around $a = 3$.

$a_0 = f(3) = 2023$

$a_1 = \frac{f'(3)}{1!} = 2$

$a_2 = \frac{f''(3)}{2!} = \frac{6}{2!} = 3$

$a_3 = \frac{f'''(3)}{3!} = \frac{24}{3!} = 4$

(b) let $f(x) = xe^x$

$f'(x) = e^x + xe^x$

$f''(x) = e^x + (e^x + xe^x) = 2e^x + xe^x$

$f'''(x) = 2e^x + (e^x + xe^x) = 3e^x + xe^x$

$f^{(4)}(x) = 3e^x + (e^x + xe^x) = 4e^x + xe^x$

$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ [power series table]

$xe^x = x\sum_{n=0}^{\infty} \frac{x^n}{n!} = x + \frac{x^2}{1!} + \frac{x^3}{2!} + \dots$

let $f(x) = T_3(x) + R_3(x)$ where $T_3(x)$: Taylor Polynomial of degree 3 for $f(x)$

$R_3(x)$: Remainder of Taylor Expansion of $f(x)$

$T_3(x) = x + \frac{x^2}{1!} + \frac{x^3}{2!}$

$R_3(x) = \frac{f^{(4)}(c)}{4!} x^4$ where c is on $(0, x)$

$0.1e^{0.1} \approx T_3(0.1) = 0.1105$

max estimate of error $= \frac{f^{(4)}(0.1)}{4!}(0.1)^4$

$= 1.666 \times 10^{-3}$

5(a) $g(x) = (1+x)(2+x^2)^{-1}$

$$g'(x) = (2+x^2)^{-1} + (1+x)(-1)(2+x^2)^{-2}(2x)$$
$$= \frac{1}{2+x^2} - \frac{2x(x+1)}{(x^2+2)^2}$$
$$= \frac{1}{x^2+2} - \frac{2x^2+2x}{(x^2+2)^2}$$

let $g'(x) > 0$

$$\frac{1}{2+x^2} - \frac{2x^2+2x}{(x^2+2)^2} > 0$$
$$(x^2+2) - (2x^2+2x) > 0$$
$$-x^2 - 2x + 2 > 0$$
$$x^2 + 2x - 2 < 0$$
$$(x+1)^2 - 3 < 0$$
$$(x+1)^2 < 3$$
$$|x+1|^2 < 3$$
$$|x+1| < \sqrt{3}$$
$$-\sqrt{3} < x+1 < \sqrt{3}$$
$$-\sqrt{3} - 1 < x < \sqrt{3} - 1$$

$\therefore$ interval where $g(x)$ is increasing $= (-\sqrt{3}-1, \sqrt{3}-1)$

(b) $$g''(x) = (-1)(x^2+2)^{-2}(2x) - \left[(4x+2)(x^2+2)^{-2} + (2x^2+2x)(-2)(x^2+2)^{-3}(2x)\right]$$
$$= -\frac{2x}{(x^2+2)^2} - \left[\frac{4x+2}{(x^2+2)^2} - \frac{8x^3+8x^2}{(x^2+2)^3}\right]$$
$$= \frac{8x^3+8x^2}{(x^2+2)^3} - \frac{6x+2}{(x^2+2)^2}$$
$$= \frac{8x^2(x+1)}{(x^2+2)^3} - \frac{2(3x+1)}{(x^2+2)^2}$$

let $g'(x) = 0$

$$\frac{1}{x^2+2} - \frac{2x^2+2x}{(x^2+2)^2} = 0$$
$$\frac{(x^2+2) - (2x^2+2x)}{(x^2+2)^2} = 0$$
$$\frac{-x^2-2x+2}{(x^2+2)^2} = 0$$
$$-x^2 - 2x + 2 = 0$$
$$x^2 + 2x - 2 = 0$$
$$(x+1)^2 - 3 = 0$$
$$(x+1)^2 = 3$$
$$x+1 = \pm\sqrt{3}$$
$$x = -1 \pm \sqrt{3}$$

$$g(-1+\sqrt{3}) = \frac{1}{4} + \frac{\sqrt{3}}{4}$$
$$g''(-1+\sqrt{3}) = -\frac{1}{4} - \frac{\sqrt{3}}{6} < 0$$

$\therefore \left(-1+\sqrt{3}, \frac{1}{4} + \frac{\sqrt{3}}{4}\right)$ is a maxima

$$g(-1-\sqrt{3}) = \frac{1}{4} - \frac{\sqrt{3}}{4}$$
$$g''(-1-\sqrt{3}) = -\frac{1}{4} + \frac{\sqrt{3}}{6} > 0$$

$\therefore \left(-1-\sqrt{3}, \frac{1}{4} - \frac{\sqrt{3}}{4}\right)$ is a minima